

Reading J -couplings from zero-field NMR spectra with a trained network

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Abstract

This is a status report on the simulation side of the project. The goal is to take one zero-field NMR spectrum and return a probability distribution over the J -couplings that produced it, rather than a single fitted number. Four molecules are done. Each trained network reaches the best precision the noise allows, and its posterior agrees to within 2.202% with an exact reference calculation (nested sampling on the true likelihood). The calibration check comes back conservative rather than overconfident, which is the safe direction. Where a coupling cannot be measured at all, the method says so instead of returning a value. Section 7 is the important caveat: when the model itself is wrong, the built-in warning fires for one of the three cases tested and stays silent for the other two, both of which shift the coupling by many times its stated error bar. That has to be fixed before real data is worth touching.

1 The problem, and why a distribution

At zero field there is no chemical shift. The spectrum is produced by the J -couplings alone, which is what makes zero-field NMR attractive for measuring them. It is also what makes the measurement hard, because the couplings are the only labels the nuclei have.

The standard way to get couplings out of a spectrum is a least-squares fit: pick starting values, minimise the mismatch, and report the curvature at the minimum as an error bar. That works well when the spectrum determines every coupling. It fails quietly when it does not, and at zero field there are directions the spectrum genuinely cannot see. Three of them are exact:

- flipping the sign of every coupling leaves the spectrum unchanged;
- exchanging two nuclei with the same gyromagnetic ratio leaves it unchanged;
- a field at angle θ_B and one at $180^\circ - \theta_B$ give identical spectra, with log-likelihoods equal to 5.187×10^{-13} in our own test.

The third one we found in this project. On top of these, the coupling between protons *inside* an equivalent group (a methyl or methylene) moves no line at all, because the selection rule $\Delta I_A = 0$ forbids it.

So the answer to “what is J ?” is not always a number. Sometimes it is “this value, this precisely”; sometimes it is “the data cannot tell you”. A posterior distribution can say both. A fitted value with an error bar can only say the first, and will say it even when it is not true.

The method in one paragraph. Simulate many spectra from known couplings. Reduce each to a short list of peak positions, amplitudes and widths, not the raw spectrum, which would be $\sim 200\,000$ points. Train a neural network to map that list back to a distribution over the parameters. Then, because our forward model is deterministic with additive Gaussian noise, the likelihood is available in closed form, so the network’s samples can be re-weighted into an *exact* posterior. The network provides the global search; the likelihood provides the accuracy.

2 Status

Item	State
Forward model	done: reproduces a published 7-spin spectrum to 7.151 mHz
Priors, summaries, training	done
Trained networks	done: four molecules, all at the precision limit
Calibration and reference	done: SBC, plus nested sampling on the exact likelihood
Comparison to least squares	done: three cases, all simulated
Behaviour when the model is wrong	done: warns in one of three cases
Figure 1: model over published data	done
Figure 2: the posterior	done
Figure 3: the calibration check	done
Figure 4: against the alternatives	done
Figure 5: real data	needs the group’s archived spectra

Everything except the last line runs on simulated data and is reproducible from the repository (204 tests, all passing).

3 Does the forward model match a real spectrum?

Before trusting anything downstream, the simulator has to reproduce a spectrum somebody actually measured. Figure 1 overlays it on the benzene- $^{13}\text{C}_1$ multiplet published by Wilzewski *et al.*

Nothing is fitted. We use the authors’ own published couplings and their own preparation sequence, and predict where the lines should fall. Of 34 measured peaks, the 13 strongest sit a mean -22.63 mHz from the nearest predicted line, with a scatter of only **7.151 mHz**.

The offset is worth a sentence, because it is easy to misread. It is *uniform*: every line is displaced by the same amount. That is the signature of reading the figure’s axis anchor slightly wrong (0.5782 pt out of 25.55 pt/Hz accounts for it), not of a physics error, which would displace lines by different amounts. Regressing the offset against frequency gives no trend. So the honest statement is: the model reproduces the multiplet to 7.151 mHz *scatter about a constant offset*, and the offset is ours, not the model’s.

4 What the trained networks achieve

Four molecules, one network each. The “floor” column is the best 95 % interval the noise permits, computed from the Fisher information of the simulator. It is a hard limit, not a target.

Molecule	Spins	95 % width	Floor	Ratio	Unmeasurable
formic acid	2	2.289 mHz	2.263	1.011	none
formaldehyde	3	1.312 mHz	1.307	1.004	J_{HH}
glycine	3	1.312 mHz	1.307	1.004	J_{HH}
methanol	4	1.038 mHz	1.012	1.025	J_{HH}

Every network lands on its own floor (ratios 1.011 to 1.025). The nuisance parameters do too: on formic acid the field magnitude, field angle and T_2 come in at 1.013, 0.985 and 1.050 times their floors.

Forward model over the published spectrum — benzene- $^{13}\text{C}_1$

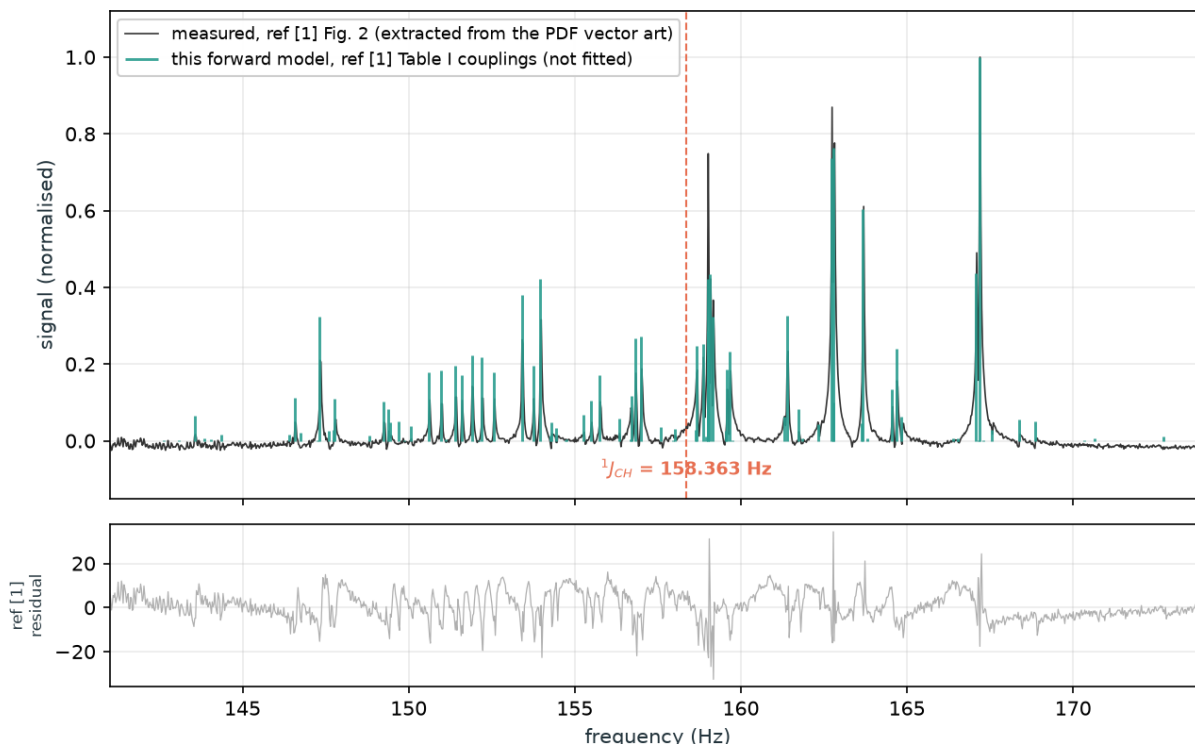


Figure 1: The forward model (no fitting) over a published spectrum. The measured trace is recovered exactly from the paper’s PDF, which is vector art, so this is the real curve rather than a digitised screenshot.

One result here is easy to miss. The precision *improves* as molecules get bigger, from 2.289 to 1.038 mHz, which is not what averaging more lines would give. The reason is in the physics: an XA_2 group puts its line at $\frac{3}{2}J$ and an XA_3 group at J and $2J$, so the lines move *faster* than J does, and the same peak-position error buys a tighter coupling. Finding this corrected a constant in our code that had only ever been right for two spins.

5 Is it calibrated?

A posterior that is confidently wrong is worse than no posterior. The standard test is simulation-based calibration: draw parameters from the prior, simulate, infer, and record where the truth falls in the posterior. Over many repeats those ranks must be uniform.

For J the outer 20 % of the range holds 0.05667 of the mass where 0.20 is expected, and the centre is correspondingly heavy. That is a conservative network, not an overconfident one, and it is exactly why the re-weighting step works well: a proposal that is too broad still covers the truth.

6 How it compares with a least-squares fit

This is the comparison that decides whether the method is worth anything, so it was done against a competent version of the alternative (restarted simplexes, normalised coordinates, and a numerical Hessian with auto-scaled steps), not a straw man. It has three different answers, and Figure 4 shows all three rather than only the flattering one.

Figure 2 — the posterior on a coupling, and its spread
 ^{13}C -formic acid, simulated spectrum, single-round NPE on 150,000 simulations, reweighted by the exact likelihood

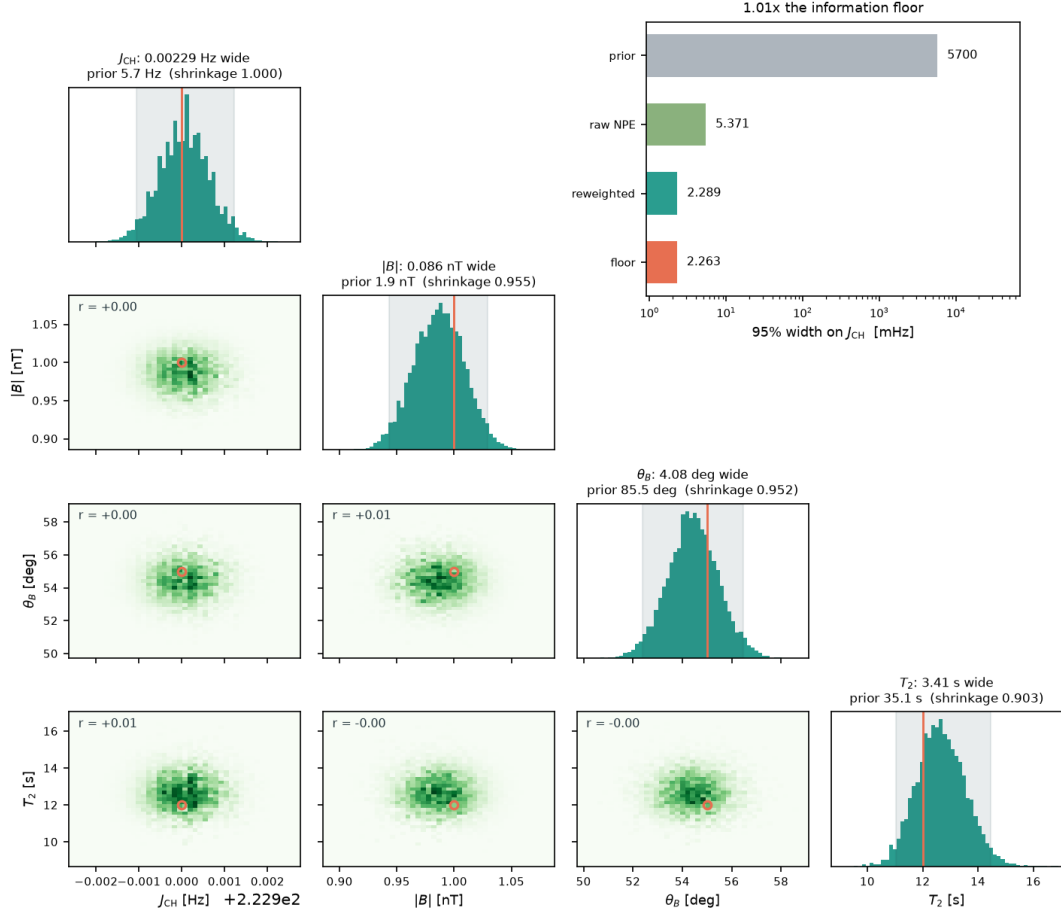


Figure 2: From a prior several Hz wide to a posterior a few mHz wide. The re-weighted posterior sits at $1.011\times$ the information floor, and the four parameters come back essentially uncorrelated.

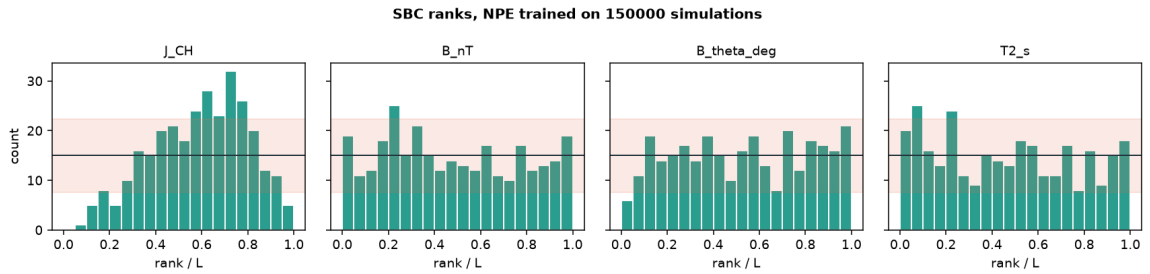


Figure 3: Calibration ranks over 300 trials. Flat is correct. For $|B|$, θ_B and T_2 they are flat. For J they are depleted at the edges, which means the posterior is too *wide*, which is the safe direction.

Figure 4 — the trained network against exact-likelihood sampling and against a least-squares fit

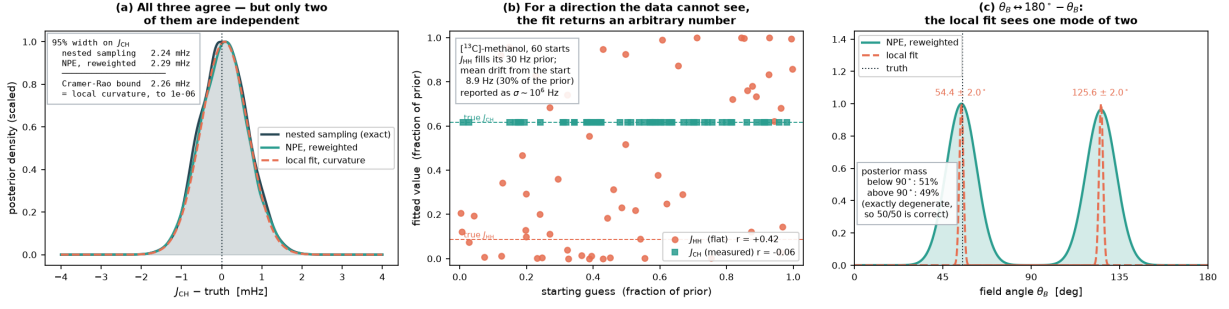


Figure 4: (a) On a clean problem all three methods agree. (b) On a flat direction the fit returns an arbitrary number with an error bar attached. (c) On a two-mode problem the fit reports one mode and excludes the other; the network keeps both.

(a) On a clean problem, they agree, and the fit is not the weak link. On formic acid all 200 random starts find the same minimum, agreeing on J to 2.022×10^{-7} Hz.

Method	95 % width on J_{CH}
Nested sampling on the exact likelihood	2.24 mHz
Trained network, re-weighted	2.289 mHz
Cramér–Rao bound (= curvature of the fit)	2.263 mHz

The rule in that table is deliberate. The least-squares error bar and the Cramér–Rao bound are not two independent results: for a Gaussian likelihood the curvature at the optimum *is* the expected information, and we measure them to agree to 1.139×10^{-6} . Listing them as separate rows would suggest a corroboration that does not exist.

What the table does say is stronger. Two fully Bayesian methods, neither of which assumes the posterior is Gaussian, land within 2.202 % of the asymptotic bound. That means the posterior *is* Gaussian to about that accuracy on this problem, which explains why the simple fit ties here, and predicts exactly where it will stop working: wherever the posterior is not Gaussian. That is cases (b) and (c).

(b) On a flat direction, the fit returns a number that means nothing. On methanol the proton–proton coupling inside the methyl group is invisible. The fit does not fail. It returns $J_{HH} = -3.542$ Hz with a 95 % interval of 1.788×10^6 Hz on a prior only 30 Hz wide, and prints that as an error bar. Across 60 starts the fitted value fills the entire prior. The network instead reports a shrinkage of 0.01086 on this parameter, which is to say: the posterior is the prior, and nothing was learned.

(c) With two modes, the fit sees one. Started at 70° the fit returns $54.4 \pm 2.0^\circ$; started at 110° it returns $125.6 \pm 2.0^\circ$. Both intervals are tight, both are honest about their own curvature, and each excludes the other. The network holds both modes, at 50.91 % and 49.09 %, which is correct, because they are exactly degenerate.

7 When the model is wrong

Everything above assumes the model is right, which on real data it will not be. The pipeline has always carried a safeguard for this: the fraction of usable samples after re-weighting is supposed

to collapse when the data cannot be explained, so the method fails loudly instead of returning a tight wrong answer. That safeguard had never been tested, because every spectrum it had ever seen came from the same simulator the inference uses.

So I built three spectra the model cannot reproduce at any parameter value, each one something that happens in a real experiment, and asked two questions of each: is the answer still right, and does anything warn you? Alongside the efficiency I measured the older and simpler check, the goodness of fit of the residuals at the best-fit point.

Figure 6 — three ways the model can be wrong, and whether the method notices

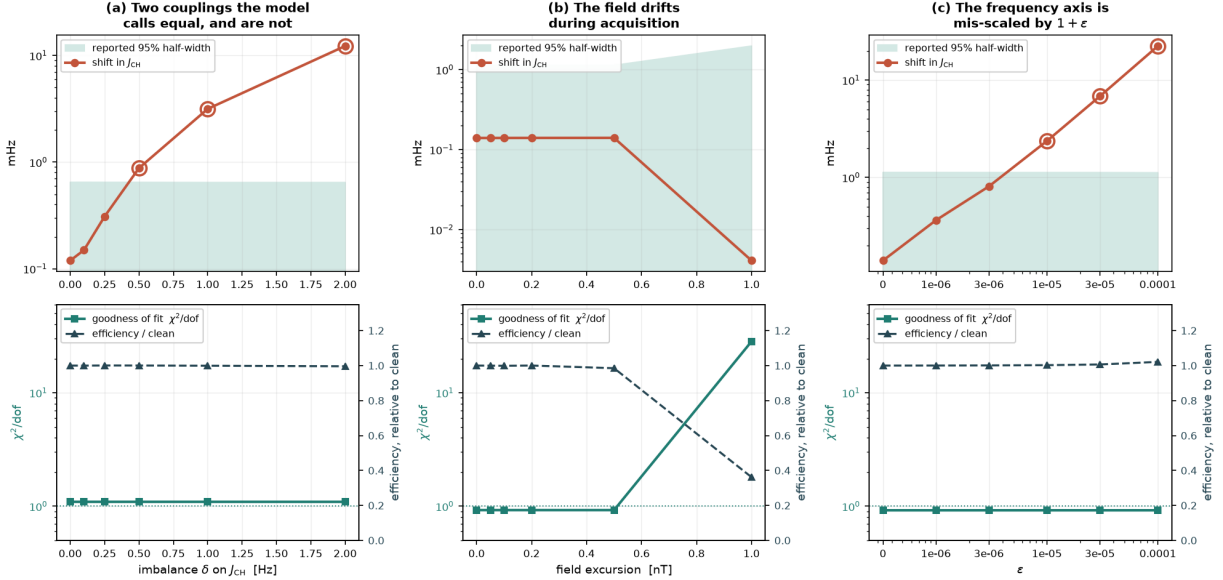


Figure 5: Top row: how far the reported coupling moves, against the error bar it is reported with. Once the point leaves the band the method is confidently wrong. Bottom row: whether either check notices. Only the middle column notices.

Two couplings the model calls equal, and are not. This is the case Wilzewski et al. invoke for their own discrepancy, where residual dipolar couplings are hard to tell from J-couplings. An imbalance of 2 Hz shifts J_{CH} by 12.27 mHz, roughly nine times its own 95 % interval, and the interval never contains the truth. Neither check reacts at all: the goodness of fit stays at 1.093 with $p = 0.4234$, and the efficiency is 0.996 times its clean value. This is a silent failure of exactly the kind the safeguard was meant to prevent.

I traced the mechanism rather than leaving it as a black box, and it is fixable. Breaking the equivalence does produce a signature: lines appear near J and $J/2$ that the selection rule forbids in the correct model, so nothing else could put them there. But they arrive four orders of magnitude weaker than the main multiplet, and the peak-list summary keeps only the strongest few lines. The evidence is discarded before the likelihood sees it, and the fit then absorbs the rest into a shifted coupling with clean residuals. Keeping a slot for lines the model does not predict would recover it.

The field drifts during acquisition. This one behaves as advertised. Below the resolution at which peaks are merged, a drift changes nothing, which is correct rather than lucky: a sub-resolution drift does not change a peak list. Above it, at 1 nT, the goodness of fit rises to 28.31 with $p = 2.324 \times 10^{-28}$ and the efficiency falls to 0.3616 of its clean value. The warning is unmissable, and the coupling stays covered because the interval widens to match. This is the behaviour the whole approach was promised on.

The frequency axis is mis-scaled. An error of $\epsilon = 1 \times 10^{-4}$ shifts J_{CH} by 22.43 mHz with the error bar unchanged, the goodness of fit unchanged at 0.9246, and the efficiency unchanged. Here the blindness is not an accident of the summary but a theorem. Multiplet lines sit at fixed multiples of J and the low-frequency line scales with the field, so scaling every frequency by $1 + \epsilon$ is reproduced exactly by $J \rightarrow J(1 + \epsilon)$, $|B| \rightarrow |B|(1 + \epsilon)$, $T_2 \rightarrow T_2/(1 + \epsilon)$. The two summaries agree to 5.82×10^{-7} of one noise unit, so there is no residual for any check to find. No amount of statistics fixes this. The frequency axis has to be calibrated independently, and the calibration uncertainty has to be carried into the result by hand.

What this means. The safeguard works on model errors that point *away* from the parameters, and is blind to those that point *along* them. That is not a detail, because the dangerous systematics are the second kind: they are absorbed into a plausible parameter value rather than showing up as a bad fit. Efficiency and goodness of fit agree with each other in all three cases, so the network’s own diagnostic adds nothing here that the classical residual test does not already give. Before this method is used on real data, either the summary has to keep the lines that carry the evidence, or the systematics have to be modelled as parameters so the posterior can widen honestly.

8 Things that did not work, or turned out otherwise

- **A bug in our own observation model, found by looking for trouble.** A peak list merges lines that are too close to resolve. We had been merging at the model’s *own* linewidth, which moves with the fitted T_2 , so a 0.1 s change in T_2 could merge three lines into one and move the log-likelihood by 5×10^5 . On real data the peak list is read off the spectrum at a fixed instrument resolution, long before T_2 is known, so the merge threshold has to be an instrument constant. It now is. Fixing it made the Hessian positive definite at the truth, brought nested sampling exactly onto the floor, and removed what had looked like multi-modality in the least-squares fit.
- **A hypothesis of ours, refuted.** We expected the sampling efficiency to drop near “resolution cliffs”, where the number of resolved peaks changes. It does not: Spearman $\rho = -0.02189$ ($p = 0.8681$) over 60 observations. Screening seven candidate explanations, only the true coupling itself survives correction for multiple testing ($\rho = -0.4072$, $p_{\text{adj}} = 0.00869$). That points at the network’s fit rather than at the physics.
- **The efficiency criterion needs re-stating.** The proposal sets a pass/fail threshold on sampling efficiency as a single number. Measured across 40 spectra from the same network and the same model, the usable fraction ranges from 0.01463 to 0.4553, a factor of 31.11. It should be read over several spectra, not one.
- **A conclusion we had to withdraw.** An earlier version of the network-tuning study appeared to show that a wider network gave the tightest proposal and the *worst* efficiency. That was an artefact of the two bugs above. After fixing them the ordering is monotonic, and the wider network is simply better (23.41 % $\rightarrow$ 37.37 % efficiency at unchanged precision).
- **Amortisation does not pay off on speed alone.** Training costs about 2537 s. A single least-squares fit takes 1.887 s. Against the multi-start protocol you would actually need for safety, the break-even point is about 6.834 spectra. The case for the network is what it reports, not how fast it reports it.

9 Where this sits in the literature

I read the five closest papers in full rather than from abstracts, and it narrowed what I should be claiming. Two points matter.

First, some things I had implied were new are not. Cobas (Mestrelab) already has a network that extracts couplings from strongly coupled spectra with no starting guess, and already pairs it with a quantum-mechanical refinement stage. Two other recent papers do report uncertainties, though what they report is the spread across several independently trained networks, which measures whether the *networks* agree, not what the *spectrum* constrains. A perfectly flat direction would give several confident, identical, wrong answers.

Second, the closest work supports the case for doing this at zero field rather than undermining it. Cobas’s four-spin paper states plainly that certain coupling combinations “cannot be determined from the spectrum alone” and resolves them by adopting an ordering convention. And the way that literature breaks the labelling degeneracy is by ranking nuclei by chemical shift, which is precisely the tool zero field does not have. So the surviving claim is narrower than “nobody has done this”, and better supported: a calibrated posterior from a single spectrum, in the one regime where the standard remedy for the degeneracy is unavailable, with flat directions reported rather than curated away.

10 Every number in this document is generated

The numbers above are not typed in. Each study script records what it owns into `results.json`; `results.py` renders that into `results_macros.tex`, which this document reads with `\input`. Re-run a study, recompile, and the text updates. `python results.py --check` exits non-zero if any document has drifted, so it can run in CI.

This was not always true, and the cost was visible in an audit: a stale table, a stale eigenvalue list, an error bar off by a factor of 1.3, and one *reversed* conclusion, all because four documents each copied the same run logs by hand.

11 What I need next

Everything above is simulated. The remaining figure is the method applied to a real spectrum, and for that I need access to the group’s archived ZULF data. Three things are essential:

1. **The spectra**, ideally raw FIDs. Which molecules does not matter (adding one is a table of couplings and a training run), so whatever is best documented is the right choice.
2. **The preparation protocol** for each dataset (thermal drop, adiabatic ramp, or pulses, with flip angles and axes). This sets the initial state, which fixes the line *amplitudes*; positions alone do not pin the couplings.
3. **Acquisition parameters**: sample rate, acquisition time, number of points.

Three more would each remove an assumption: the resolution of the peak-fitting step (see the merge bug above; I would rather have the instrument’s number than a guess), the residual field if it was logged, and the magnetometer bandwidth and any hardware filter. Any previously fitted couplings would let me compare against the existing fitter on the same data rather than only against simulation.

Even one or two well-documented datasets would be enough to make the last figure real.